

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech

Semester: V

Name of the paper: Renewable Energy

Paper Code: TME 509

Time: 3 hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1.	20 marks	
(a)	Compare renewable and fossil energy sources in terms of resource availability, capacity factor, and emissions. Discuss the key challenges and barriers to large-scale renewable energy integration into power systems.	CO 1/CO2
(b)	With a labeled diagram, explain a flat plate solar collector: absorber, glazing, insulation, flow arrangements. Discuss thermal losses and methods to enhance performance (selective coatings, flow rate control).	
(c)	Explain solar cookers: box-type and concentrating cookers. Discuss optical/thermal design features, cooking power, safety, and practical deployment issues.	
Q2.	20 marks	
(a)	Compare and contrast aerobic and anaerobic digestion processes in biomass conversion. Describe the key factors affecting the efficiency of anaerobic digestion.	CO 2/CO3/CO4
(b)	Discuss the biogas yield and combustion characteristics of typical biomass feedstocks. Explain the utilization of biogas for cooking and other bioenergy applications in domestic and industrial sectors.	
(c)	Compare biochemical and thermochemical conversion routes for biomass. Detail products, operating conditions, scales, advantages/limitations, and suitable feedstocks.	
Q3.	20 marks	
(a)	What are the major criteria for site selection of a wind power plant? Explain the main components and working of a wind turbine generator system with a neat sketch.	CO2/CO4/CO5
(b)	Explain the concept and working of a solar desalination system (solar still). Discuss factors affecting its performance and methods to enhance water yield.	
(c)	Describe the methods of harnessing geothermal energy. Explain with neat sketches the working of dry steam, flash steam, and binary cycle geothermal power plants.	
Q4.	20 marks	
(a)	Explain the different types of geothermal resources based on geological conditions and temperature levels. Discuss their formation, characteristics, and suitability for various energy applications such as electricity generation and direct heating.	CO4/CO5
(b)	Explain the principles of tidal energy conversion. Discuss the types of tidal power plants, their working, and limitations.	
(c)	Describe the principle of operation of Ocean Thermal Energy Conversion (OTEC). Explain how the temperature difference between surface and deep ocean water is utilized for power generation.	
Q5.	20 marks	
(a)	Explain the principles of energy conservation. Discuss the importance of efficient energy use in domestic and industrial sectors.	CO 5/CO 6
(b)	Describe the concepts of converting wave energy into mechanical or electrical energy.	
(c)	Discuss the types and development of OTEC plants closed cycle, open cycle, and hybrid systems. Explain their working, advantages, and limitations.	

